Operating Systems

Banana – Unit 4

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Class: 4 K

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1A.

```
arjun@arjun-VirtualBox:~/Desktop/os-banana4/PES1UG24CS922-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
```

1B.

```
arjun@arjun-VirtualBox:~/Desktop/os-banana4/PES1UG24CS922-pes-vcs$ find .pes/objects -type f
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
```

2A.

```
arjun@arjun-VirtualBox:~/Desktop/os-banana4/PES1UG24CS922-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
```

2B.

```
arjun@arjun-VirtualBox:~/Desktop/os-banana4/PES1UG24CS922-pes-vcs$ xxd .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12 | head -20
00000000: 626c 6f62 2031 3600 4865 6c6c 6f2c 2050  blob 16.Hello, P
00000010: 4553 2d56 4353 210a                      ES-VCS!.
```

3A.

```
arjun@arjun-VirtualBox:~/Desktop/os-banana4/PES1UG24CS922-pes-vcs$ cat .pes/index
304444 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776320977 6 a.txt
304444 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776321056 6 file1.txt
304444 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776321056 6 file2.txt
arjun@arjun-VirtualBox:~/Desktop/os-banana4/PES1UG24CS922-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
arjun@arjun-VirtualBox:~/Desktop/os-banana4/PES1UG24CS922-pes-vcs$ ./pes add file1.txt file2.txt
arjun@arjun-VirtualBox:~/Desktop/os-banana4/PES1UG24CS922-pes-vcs$ ./pes status
Staged changes:
  staged:     a.txt
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  .gitignore
  untracked:  tree.h
  untracked:  pes.c
  untracked:  Makefile
  untracked:  pes.h
  untracked:  commit.h
  untracked:  index.h
  untracked:  test_sequence.sh
  untracked:  tree.c
  untracked:  object.c
  untracked:  test_tree.c
  untracked:  README.md
  untracked:  test_objects.c
  untracked:  index.c
  untracked:  commit.c
```

3B.

```
arjun@arjun-VirtualBox:~/Desktop/os-banana4/PES1UG24CS922-pes-vcs$ cat .pes/index
304444 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776320977 6 a.txt
304444 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776321056 6 file1.txt
304444 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776321056 6 file2.txt
```

4A.

```
arjun@arjun-VirtualBox:~/Desktop/os-banana4/PES1UG24CS922-pes-vcs$ ./pes log
commit ab4886a5049346525e19fe773e0ac9a69c23a374925138425627de724b2f14d4
Author: PES User
Date:   0



commit 8eabb82edad72985d1a4932fbf20cf6b31b5aa764f8e2de316805ab11502f290
Author: PES User
Date:   0


Add farewell


commit 98cb2a69c59d421c3ff74125f26cb72368695f50b1f0f79006db189bd33e1ead
Author: PES User
Date:   0
```

4B.

```
arjun@arjun-VirtualBox:~/Desktop/os-banana4/PES1UG24CS922-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/6e/f19b41225c5369f1c104d45d8d85efa9b057b53b14b4b9b939dd74decc5321
.pes/objects/8e/abb82edad72985d1a4932fbf20cf6b31b5aa764f8e2de316805ab11502f290
.pes/objects/98/cb2a69c59d421c3ff74125f26cb72368695f50b1f0f79006db189bd33e1ead
.pes/objects/ab/4886a5049346525e19fe773e0ac9a69c23a374925138425627de724b2f14d4
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/refs/heads/main
```

4C.

```
arjun@arjun-VirtualBox:~/Desktop/os-banana4/PES1UG24CS922-pes-vcs$ cat .pes/refs/heads/main
ab4886a5049346525e19fe773e0ac9a69c23a374925138425627de724b2f14d4
arjun@arjun-VirtualBox:~/Desktop/os-banana4/PES1UG24CS922-pes-vcs$ cat .pes/HEAD
ref: refs/heads/main
```

Q5.1 **pes checkout <branch>**

- Read commit hash from .pes/refs/heads/<branch>
- Update .pes/HEAD to:

ref: refs/heads/<branch>

- Load commit → get tree → recreate working directory:
    - Write blobs as files
    - Create directories recursively
    - Remove files not in target tree
- Update index to match tree

Complexity:

- Must safely overwrite working directory
- Handle file changes, deletions, directories
- Prevent data loss (uncommitted changes)

Q5.2 **Detect dirty working directory**

- For each index entry:
    - stat() file → compare mtime and size

    - If different → file modified
- Load target branch tree and compare hashes

Conflict if:

- File modified locally AND
- File exists in target branch AND
- Hash differs

Then abort checkout

Q5.3 **Detached HEAD**

- HEAD points directly to commit, not branch
- New commits are created but not referenced by any branch

Effect:

- Commits become unreachable

Recovery:

pes branch <name> <commit_hash>

Q6.1: Garbage collection

Algorithm:

1. Start from all branch heads
2. Traverse:

commit → tree → blobs

3. Store visited hashes in a **hash set**
4. Delete objects not in set

Estimation:
~100k–300k objects visited (due to shared history)

Q6.2: GC race condition

Problem:

- Commit writes objects but hasn't updated HEAD yet
- GC runs → sees objects as unreachable → deletes them

Result:

- Broken commit (missing objects)

Git avoids this by:

- Atomic writes (write + fsync + rename)
- Running GC separately (not concurrently with commits)